

Power Functions

Date _____ Period _____

Consider each power function. Determine the power and constant of variation.

1) $f(x) = 6x^{\frac{3}{7}}$

Power: $\frac{3}{7}$ Constant: 6

2) $f(x) = 5x^{-6}$

Power: -6 Constant: 5

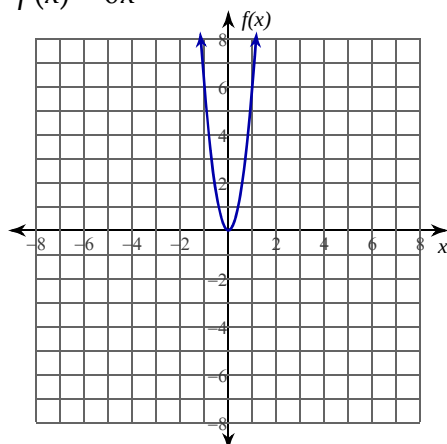
3) $f(x) = 8x^{-\frac{1}{3}}$

Power: $-\frac{1}{3}$ Constant: 8

4) $f(x) = 2x^{\frac{5}{7}}$

Power: $\frac{5}{7}$ Constant: 2**Consider each power function. Determine the domain and range, intercepts, end behavior, continuity, and regions of increase and decrease.**

5) $f(x) = 6x^2$

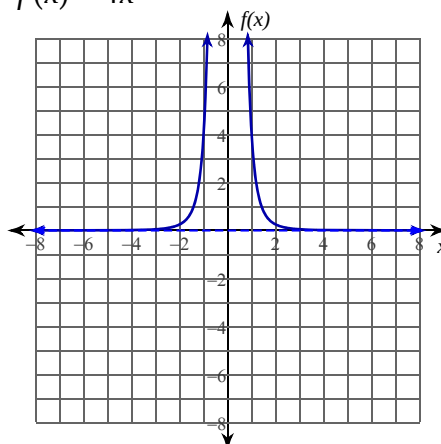
Domain: $(-\infty, \infty)$ Range: $[0, \infty)$

x-intercept: 0 y-intercept: 0

$$\lim_{x \rightarrow -\infty} f(x) = \infty \quad \lim_{x \rightarrow \infty} f(x) = \infty$$

Continuous on $(-\infty, \infty)$ Increasing: $(0, \infty)$ Decreasing: $(-\infty, 0)$

6) $f(x) = 4x^{-4}$

Domain: $(-\infty, 0) \cup (0, \infty)$ Range: $(0, \infty)$

No intercepts

$$\lim_{x \rightarrow -\infty} f(x) = 0 \quad \lim_{x \rightarrow \infty} f(x) = 0$$

Infinite discontinuity at $x = 0$ Increasing: $(-\infty, 0)$ Decreasing: $(0, \infty)$

7) $f(x) = 5x^{-4}$

Domain: $(-\infty, 0) \cup (0, \infty)$ Range: $(0, \infty)$

No intercepts

$$\lim_{x \rightarrow -\infty} f(x) = 0 \quad \lim_{x \rightarrow \infty} f(x) = 0$$

Infinite discontinuity at $x = 0$ Increasing: $(-\infty, 0)$ Decreasing: $(0, \infty)$

8) $f(x) = 8x^{-\frac{4}{3}}$

Domain: $(-\infty, 0) \cup (0, \infty)$ Range: $(0, \infty)$

No intercepts

$$\lim_{x \rightarrow -\infty} f(x) = 0 \quad \lim_{x \rightarrow \infty} f(x) = 0$$

Infinite discontinuity at $x = 0$ Increasing: $(-\infty, 0)$ Decreasing: $(0, \infty)$